Arnab Mandal

+91-9560452773 | arnabmandal2912@gmail.com | linkedin.com/arnabmandal2912 | github.com/arnabwithab | Portfolio

INTERNSHIPS & EXPERIENCE

Intern, Enterprise AI Platform, Fidelity International, Gurgaon Jun 2026 - Jul 2026

- Slashed short-context **PII** redaction latency to sub-**100ms** by serving fine-tuned GLiNER on Ray at **100**% F1.
- Scaled long-context PII redaction by fine-tuning and quantizing Gemma 4 cutting latency to **300ms** at **96**% F1.
- Shipped document sensitivity guardrail layer for LiteLLM on **15k**+ monthly uploads, audited via **ELK**.
- Built a DeepEval **harness** for RAG evaluation, tuning reranking and search to achieve **97%** faithfulness.

Freelance AI Engineer, value-based care healthcare client Feb 2026 - Apr 2026

- Deployed a multi-agent Text2SQL engine with sandboxed-agent execution serving **21** orgs and **200**+ users on AWS.
- Achieved **p95**<**5s** via tiered agent routing, auto-scaling Fargate, and semantic caching at **50**+ concurrent users.
- Built HNSW-indexed pgvector Schema GraphRAG layer, achieving **0.87** context recall over domain knowledge.
- Halved LLM calls and raised execution accuracy from 73% to **94%** by denormalizing data into dbt marts.

AI Engineer Intern, DeepLure Research Pvt Ltd, Amravati Dec 2025 - Jan 2026

- Optimized inference service for **50**+ clients reducing per-request latency from 30ms to **2ms**.
- Migrated to TensorRT + Triton, achieving **8x** throughput, reducing AWS GPU costs by **$2,300**+/month.

Founding AI Intern, Critical AI Pvt Ltd, Bengaluru Jul 2025 - Sep 2025

- Deployed an on-prem RAG agent for a legal firm, cutting inference cost **30%** vs. cloud for **early adoption** of AI.
- Ran live demo for a voice multimodal RAG agent for geospatial queries, pitched to Indian Navy departments.

AI Intern, WESEE, Ministry of Defence, New Delhi May 2025 - Jul 2025

- Built LLM serving for internal MoD chatbot, used by **7**+ teams and handling **200**+ daily queries.

PROJECTS

Open-Source Contributor at vLLM: Domino Speculative Decoding

- Implemented GRU-corrected parallel speculative decoding, reporting a **45%** throughput gain on Qwen3-8B.
- Improved acceptance length by **68%** over DFlash across **7** benchmarks, including GSM8K and HumanEval.
- Extended Inference and Distributed training support for Domino across vllm & vllm-speculators.

AutoLinks: Distributed Semantic Internal Linking Engine for SEO

- Designed distributed Go service that recommends internal links for SEO using NER & vector search on AWS.
- Architected Goroutine workers with ElastiCache Streams for sitemap ingestion, load-tested to **150** articles/min.
- Achieved **600 ms** p95 latency per recommendation, at a **85%** top-5 link acceptance rate per LLM judge offline eval.

Open-source contributions across LLM inference & AI agent infrastructure

- **llm-d**: enabled KEDA/HPA autoscalers to scale zero-instance inference, via new Prometheus metric.
- **OpenMetadata**: Added provenance-based filters to AI context layer, against stale context, inactive knowledge, etc.
- **NVIDIA-NeMo**: Preserved parent-agent context in synchronous wrappers and fixed prompt evaluation failures.
- **WebMCP**: Fixed WebMCP compatibility between native chrome & stub dev, for headless web agents.

EDUCATION

Shiv Nadar University Noida, Delhi-NCR, India
B.Tech in Computer Science Engineering **(8.59/10.0)** *Aug 2023 - Jul 2027*

TECHNICAL SKILLS

Languages	Python, Go, TypeScript, CUDA, SQL
AI & ML	LangChain, LangGraph, LangSmith, LiteLLM, Optuna, MLflow
MLOps & Infra	vLLM, Triton, TensorRT, Ray Serve, Kubernetes, Docker, Jenkins, Prometheus
Cloud	GCP, AWS, Azure, Terraform

ACHIEVEMENTS

- **Smart SNU Hackathon'25 Winner**: Led a 6-member team to build and deploy AI solutions for farmers.
- **LeetCode Knight**: Achieved a **1,925 peak contest rating** and **solved 500+ DSA problems.**
- Semifinalist speaker at SMVIT Parliamentary Debate'25